

# Syllabus

## Spring semester 2026

| Course Title | General Physics 2 |           | Course Code |              | PE3103          |                    | Credit | 4 | Classroom | Hyung 3303 |
|--------------|-------------------|-----------|-------------|--------------|-----------------|--------------------|--------|---|-----------|------------|
|              |                   |           | Type        | 자연계열/교과심화    | Hours           | 4                  |        |   |           |            |
| Teacher      | Name              | Class no. | Office      | Office Hour* | Contact         | E-mail             |        |   |           |            |
|              | Lee, Hun Bong     | 1         | Chang 5505  |              | Anytime(e-mail) | superpig@ksa.hs.kr |        |   |           |            |
|              |                   |           |             |              |                 |                    |        |   |           |            |
|              |                   |           |             |              |                 |                    |        |   |           |            |
|              |                   |           |             |              |                 |                    |        |   |           |            |

\*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

### 1. Course Descriptions

General Physics 2 is one-semester course which covers Electricity and Magnetism, Electric Circuits, Electromagnetic Waves and Wave Optics, Relativity and Quantum theory. The course is a university level course providing a deep understanding of the concepts and principles of physics and emphasizing the creative thinking. The course is offered for students who have completed General Physics I course and who have completed or are enrolled concurrently in a differential and integral calculus course. A systematic approach to problem solving is considered as a basic way to get better understanding. Questions and problems also develop students' mathematical skill.

### 2. Core Target Competencies

| Subject Core Competencies                                                                                                                                                                                                                                                                                                                                                                                      |                                  |                                |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------|------------|
| Critical Thinking Skills                                                                                                                                                                                                                                                                                                                                                                                       | Analytical and Creative Thinking | Complex Problem-Solving Skills | Creativity |
| Fundamental concepts in several subjects in physics will be presented, including electromagnetism, optics, and subjects in modern physics, are introduced. Following the homework and the discussion sessions in this lecture, students will be encouraged to develop their competencies listed above. The ways of thinking, trained in this lecture, will provide important background for future coursework. |                                  |                                |            |

### 3. Textbooks and References

- A. Textbook: University Physics with Modern Physics, 2nd Ed. (Bauer, Westfall, McGraw-Hill)
- B. References
- The Feynman Lectures on Physics I, II, III (R. Feynman et al., Addison-Wesley)
  - Berkeley Physics Course I (McGraw-Hill)

### 4. Class Method

- A. Self-learning during homework, assessment through short quizzes.
- B. The lectures will provide students with various approaches to the subject that differ from the textbook.
- C. Wide use of physical demonstrations and computer simulations.

※

## 5. Evaluation

### A. Evaluation Criteria

| Midterm | Final | Quizzes | Homework | Attendance and Attitude | Total |
|---------|-------|---------|----------|-------------------------|-------|
| 33      | 34    | 0       | 17       | 16                      | 100   |

### B. Evaluation Method

- (1) The midterm and final exams are given through written tests.-
- (2) Quizzes will be given several times during the semester.-
- (3) Homework: Penalty will be given to the overdue homework.-
- (4) Attendance: -0.3 point per late & -1.0 point per absence without notice.

-

## 6. Criteria for Awarding Credits

☞ Pass: In the case of receiving a **score of 80 or more** out of 100 according to the criteria of the evaluation method.

☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- 
- Calculation of the basis score for awarding credit to absentees :
- 
- If a basis score cannot be calculated, the grading method :
- 
- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

## 7. Weekly Plans

| Week (period)                | Topic and Contents                                                                                     | (Notes)                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Week 1</b><br>2.23.~3.1.  | Ch.21 Electrostatics<br>Ch.22 Electric Fields and Gauss's Law                                          | Report would be represented after finishing each chapter |
| Core Competencies            | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                                                          |
| <b>Week 2</b><br>3.2.~3.8.   | Ch.23 Electric Potential                                                                               | <b>3.2.</b><br><b>Substitute Holiday</b>                 |
| Core Competencies            | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                                                          |
| <b>Week 3</b><br>3.9.~3.15.  | Ch.24 Capacitors                                                                                       |                                                          |
| Core Competencies            | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                                                          |
| <b>Week 4</b><br>3.16.~3.22. | Ch.25 Current and Resistance<br>Ch.26 Direct Current Circuits                                          |                                                          |
| Core Competencies            | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                                                          |
| <b>Week 5</b><br>3.23.~3.29. | Ch.27 Magnetism<br>Ch.28 Magnetic Fields of Moving Charges                                             |                                                          |
| Core Competencies            | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                                                          |
| <b>Week 6</b><br>3.30.~4.5.  | Ch.29 Electromagnetic Induction                                                                        |                                                          |
| Core Competencies            | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                                                          |
| <b>Week 7</b><br>4.6.~4.12.  | Ch.30 Alternating Current Circuits                                                                     |                                                          |
| Core Competencies            | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                                                          |
| <b>Week 8</b><br>4.13.~4.19. | Midterm Exams (4.14.~4.17.)                                                                            | <b>4.13.</b><br><b>Regular Classes</b>                   |

|                               |                                                                                                        |                           |
|-------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Week 9</b><br>4.20.~4.26.  | Ch.31 Electromagnetic Waves                                                                            |                           |
| <b>Week 10</b><br>4.27.~5.3.  | Ch.34 Wave Optics                                                                                      |                           |
| <b>Core Competencies</b>      | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                           |
| <b>Week 11</b><br>5.4.~5.10.  | Ch.35 Relativity                                                                                       | <b>5.5.</b>               |
| <b>Core Competencies</b>      | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity | <b>Children's Day</b>     |
| <b>Week 12</b><br>5.11.~5.17. | Ch.36 Quantum Physics                                                                                  |                           |
| <b>Core Competencies</b>      | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                           |
| <b>Week 13</b><br>5.18.~5.24. | Ch.37 Quantum Mechanics                                                                                |                           |
| <b>Core Competencies</b>      | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity |                           |
| <b>Week 14</b><br>5.25.~5.31. | Ch.38 Atomic Physics                                                                                   | <b>5.25.</b>              |
| <b>Core Competencies</b>      | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity | <b>Substitute Holiday</b> |
| <b>Week 15</b><br>6.1.~6.7.   | Week for a missing lecture                                                                             | <b>6.3.</b>               |
| <b>Core Competencies</b>      | Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity | <b>Local Election Day</b> |
| <b>Week 16</b><br>6.8.~6.14.  | <b>Make-up class week</b>                                                                              |                           |
| <b>Week 17</b><br>6.15.~6.19. | <b>Final Exams (6.18.~6.19.)</b>                                                                       |                           |

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

| 1. Analytical and Creative Thinking                                                          | 2. Complex Communication                                                     | 3. Leadership and Teamwork                                                                                       | 4. Digital/ Quantitative Literacy                                                                                                                                                                       | 5. Global Perspective                                                                       | 6. Adaptability/ Progression / Spirit of Adventure                                                                   | 7. Integrity and Ethical Decision Making                                                                     | 8 Habits/ Mindset                |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------|
| Ability to identify, manage and deal with complex problems                                   | Ability to understand and express themselves in two or more languages        | Ability to be the first to come up with and present new ideas                                                    | Ability to understand, use and apply digital technology                                                                                                                                                 | Ability to develop an open mind about the traditions or values of others                    | Ability to develop flexibility, agility and adaptability                                                             | Ability to sustain an empathetic and understanding perspective                                               | Sincerity                        |
| Ability to see through prejudice and identify reliable information and uncertain information | Ability to communicate clearly with diverse audiences or readers             | Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority | Ability to create digital contents and media                                                                                                                                                            | Ability to understand the history, politics, religion, and culture of another region        | Ability to challenge unfamiliar situations                                                                           | Ability to cultivate sincerity, honesty, fairness, and respect creativity                                    | Creativity                       |
| Ability to find quality information in the flood of information                              | Ability to listen attentively                                                | Ability to build trust, resolve conflicts, and support others                                                    | Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats                                                                                                  | Ability to use one or more of the international languages used worldwide                    | Ability to explore and experiment                                                                                    | Ability to display moral courage in the face of injustice                                                    | A love of learning and curiosity |
| Ability to articulate meaningful questions                                                   | Ability to speak effectively and coherently                                  | Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes                  | Ability to master and use higher mathematics                                                                                                                                                            | Ability to use technology to connect people and things around the world                     | Ability to word effectively in situations of ambiguous or changing priorities                                        | Ability to act responsibly for the benefit and happiness of the greater community                            | Resilience/elastcity             |
| Ability to analyze and create ideas or knowledge                                             | Ability to write concisely and clearly, considering the diversity of readers | Ability to teach, guide, and counsel<br>Ability to seek help from others                                         | Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology | Social and intellectual skills to effectively navigate information across multiple cultures | Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes | Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies | Perseverance/patience            |
| Ability to devise and test solutions to problems by                                          | Ability to explain information and persuasively                              | Ability to collaborate on tasks, manage groups, and                                                              |                                                                                                                                                                                                         | Ability to use 21 <sup>st</sup> century core competencies such as communication             | Ability to develop a sense of independence to seek new roles,                                                        | Ability to make rational and ethical decisions on complex                                                    | Self-efficacy                    |

| 1. Analytical and Creative Thinking                                                                      | 2. Complex Communication         | 3. Leadership and Teamwork           | 4. Digital/ Quantitative Literacy | 5. Global Perspective                                                                                                                       | 6. Adaptability/ Progression / Spirit of Adventure                                                 | 7. Integrity and Ethical Decision Making | 8 Habits/ Mindset        |
|----------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------|
| trial and error                                                                                          | engage others in an engaging way | delegate responsibilities            |                                   | skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues                       | ideas, or strategies                                                                               | issues                                   |                          |
| Ability to think of different ways to cope                                                               |                                  | Ability to listen and practice goals |                                   | Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue | Ability to develop the knowledge and competencies necessary to start a business and run a business |                                          | Stress management skills |
| Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields |                                  | Ability to share credit for success  |                                   | Ability to generate diverse ideas and exploit social and cultural differences to get things done                                            |                                                                                                    |                                          | Time management skills   |
| Ability to make sustained inferences                                                                     |                                  |                                      |                                   |                                                                                                                                             |                                                                                                    |                                          |                          |
| Ability to put things together or adapt them to new uses and situations                                  |                                  |                                      |                                   |                                                                                                                                             |                                                                                                    |                                          |                          |
| Ability to solve new problems that cannot be solved through established rules                            |                                  |                                      |                                   |                                                                                                                                             |                                                                                                    |                                          |                          |
| Ability to use knowledge or creativity to solve complex real-world problems                              |                                  |                                      |                                   |                                                                                                                                             |                                                                                                    |                                          |                          |

|                                                |                                                                                                                                                                            |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Complex Problem-Solving Skills</b>       | Ability to solve new and unclear problems that may arise in complex real-world situations                                                                                  |
| <b>2. Critical Thinking Skills</b>             | Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning                                |
| <b>3. Creativity</b>                           | Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem                            |
| <b>4. Human Resource Management Skills</b>     | Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, and systematic efficient management |
| <b>5. Ability to Collaborate</b>               | Ability to adjust one's own behavior in response to the behavior of others                                                                                                 |
| <b>6. Emotional Ability</b>                    | Ability to perceive other people's reactions and understand why they react the way they do                                                                                 |
| <b>7. Judgement and Decision-making Skills</b> | Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem                                            |
| <b>8. Orientated Towards Service</b>           | Ability to actively seek ways to help others                                                                                                                               |
| <b>9. Negotiation Skills</b>                   | Ability to bring together parties who differ in ideas or interests and try to reduce those differences                                                                     |
| <b>10. Cognitive Flexibility</b>               | Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways                                                               |